

Ved Patel
Course : **Bachelor of Engineering, Mechanical Engineering, 2028**
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CGPA : 8.49/10

ACADEMIC DETAILS

COURSE	SPECIALISATION	INSTITUTE	BOARD	SCORE	YEAR
CLASS XII	SCIENCE/MATHS	Green Valley High School, Vadodara	CBSE	93.8%	2023-24
CLASS X	GENERAL	Navrachana School Sama, Vadodara	CBSE	94.2%	2021-22

CORE AND ELECTIVE COURSES

Core Courses: Advanced Mechanics of Solids, Fluid Mechanics, Applied Thermodynamics, Material Science and Engineering, Heat Transfer, Design of Machine Elements, Advanced Manufacturing Processes, Mechanisms and Machines, Manufacturing Management, Engines Motors and Mobility

Electives(including Robotics Minor)**:** Robotics, Artificial Intelligence for Robotics, Finite Element Method, Internet of Things

Humanity Electives: Science Technology and Modernity, Cinematic Adaptation

INTERNSHIPS AND PROFESSIONAL EXPERIENCE

Summer Intern - DRDO CEMILAC, Bengaluru — May 2026 - July 2026

Centre for Military Airworthiness and Certification, a premier aerospace research and certification laboratory under DRDO

- Developed a **2D incompressible fluid flow solver** from scratch in MATLAB using the **Finite Difference Method** for aerodynamic modeling.
- Bypassed commercial CFD software to save **₹50L+** in licensing costs while retaining absolute control over all simulation parameters
- Successfully captured, simulated, and analyzed complex transient aerodynamic flow phenomena, such as **von Kármán vortex shedding**
- Completed a comprehensive, **expert-led training** course on **CFD++** simulation software directly from the company's representatives

Design Core, Radio Control Club — January 2025 - Present

BITS Pilani's premier aeromodelling club dedicated to the engineering, fabrication, and flight-testing of custom unmanned aircraft

- Engineered scale RC aircraft **(B-2, F-22)** and optimized custom airfoils using **XFLR5** to simulate diverse structural geometries
- Fabricated lightweight airframes using **carbon fiber** and epoxy layups, alongside composite builds of Depron, foam, and balsa wood
- Integrated flight controllers and autonav systems, mastering complex multi-axis RC flight dynamics via rigorous **PicaSim** simulations

POSITION OF RESPONSIBILITY

System Engineer, Mechanical Subsystem, CRISS Robotics — February 2026 - Present

BITS Pilani's student-led engineering team developing autonomous Mars rovers for extreme environments and international competitions

- Engineered and assembled rugged mechanical frameworks and chassis for an autonomous Mars rover prototype using **SolidWorks** and **Fusion 360**
- Validated structural durability, fatigue life, and load capacity through rigorous **ANSYS simulations** for high-stress and extreme environments
- Helped secure 2nd in Asia at **ERC** 2025, a Top 10 finish at the **International Rover Challenge**, and being the only Indian team invited to **ARCh** 2026

Coordinator, PCrA — April 2026 - Present

Department of Publications and Correspondence for BITS Pilani's technical festival, APOGEE

- Spearheaded external **correspondence** and strategic inter-college outreach campaigns, driving a record external footfall of **1,000+ students**
- Conducted a college-wide club photography and printing drive, raising **₹25L+ in revenue** to significantly fund the technical fest budget
- Managed campus-wide **publication** requests, press releases, and print promotional material distribution across all organizing departments

Treasurer, Gurjari Association — April 2026 - Present

The Gujarati cultural association for the campus, hosting Garba Nite, a student-run event and the largest cultural event by an association on campus

- Directed a **₹2L event budget** and **independently** managed all financial operations, vendor transactions, and cash flows for the annual Garba Nite
- **Generated ₹3L+** in gross revenue by executing strategic ticketing sales campaigns and securing high-value corporate **sponsorships**
- Successfully orchestrated event logistics, vendor coordination, and hospitality operations, hosting a record footfall of **3,000+ attendees**

VOLUNTEERING AND ACHIEVEMENTS

Umang Head, National Service Scheme (NSS)

Orchestrated donation drives across students, faculty, and alumni, raising **₹16L+** over two years to sponsor **150+ underprivileged students.**

School Prefect, Navrachana School

Served as student **council leader**, maintaining discipline and coordinating large-scale events while liaising between students and administration.

SKILLS

SKILLS: Fusion 360, Solidworks, ANSYS, CFD++, XFLR5, MATLAB, Simulink, C Programming, Python, Arduino IDE, Bamboo Studio, Orca, AutoCAD